

Abhishek Kumar Mishra

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EDUCATION

Lovely Professional University

B.Tech in Computer Science and Engineering | CGPA: 8.31

Phagwara, Punjab

Aug 2023 – Present

LA Garden High School

Intermediate (PCM) | 78.83%

Ranchi, Jharkhand

Apr 2021 – Mar 2022

WORK EXPERIENCE

Full Stack Developer

NGO Arya Bhumi Seva Sansthan

Remote

Dec 2023 – Sep 2025

- Architected the backend for a Blood Donation Management platform, designing normalized database schemas and query logic to search and filter donors by blood group and location, and built RESTful APIs integrating Razorpay payment workflows for the NGO's donation website.
- Automated AWS EC2 deployment through Git-based CI workflows and built a role-based admin dashboard backend with access-control logic for managing volunteers and events.

PROJECTS

SplitMate – Expense-Sharing Platform with Custom Split & Reconciliation Engine | [GitHub](#)

Jun 2026

- Built a ledger-style backend where balances are always derived from immutable Expense/Settlement records (never stored/mutated), supporting exact, percentage, and share-based splits plus a temporal-membership model that blocks users from splits outside their join/leave window.
- Integrated Razorpay for online settlements: server-side order creation and HMAC-SHA256 payment signature verification before committing a settlement, with idempotency guards on critical write endpoints to safely handle duplicate/retried requests.
- Designed a 3-step CSV import pipeline running 19 distinct anomaly-detection checks (duplicate rows, casing mismatches, settlements misclassified as expenses) with mandatory review before database writes.

Tech Stack: Node.js, Express, PostgreSQL, Prisma, Razorpay, JWT, REST APIs

RateShield – Standalone API Rate Limiting Service | [GitHub](#)

May 2026

- Engineered a rate-limiting service using a Strategy + Factory pattern to support pluggable Token Bucket and Sliding Window algorithms behind a unified API.
- Implemented atomic rate-limit checks with Redis-backed Lua scripts to eliminate race conditions under concurrent load, ensuring consistent allow/deny decisions at scale.
- Built a 55+ test unit/integration suite plus load tests targeting sub-100ms p95 latency; containerized and automated via CI.

Tech Stack: Node.js, Redis, PostgreSQL, Docker, GitHub Actions, Load Testing

Banking System – Multi-Threaded C++ Transaction Engine with TCP Server | [GitHub](#)

Apr 2026

- Designed a modular C++20 banking engine (inheritance, polymorphism, encapsulation) with a custom thread pool and thread-safe transaction queue (`std::condition_variable`, `std::future`) to process concurrent deposits, withdrawals, and transfers asynchronously.
- Applied fine-grained locking (`std::shared_mutex`, `std::scoped_lock`) across account objects to guarantee thread safety and deadlock-free multi-account transfers, exposed via a cross-platform TCP server (POSIX/Winsock) supporting live socket-based DEPOSIT/WITHDRAW/TRANSFER/BALANCE operations.
- Benchmarked throughput via synthetic workload generation, scaling from 317K to 657K txns/sec across 1–8 worker threads while tracking p50/p95/p99 latency; persisted to SQLite and validated correctness with GoogleTest.

Tech Stack: C++20, Multithreading, TCP/IP Sockets, CMake, SQLite, GoogleTest, POSIX

TECHNICAL SKILLS

Languages: C++ (C++20), JavaScript, TypeScript, SQL

Data Structures & Algorithms: Arrays, Graphs, Trees, DP, Hashing – 800+ problems solved, 1694 LeetCode Contest Rating

Systems & Concurrency: Multi-threading, Thread Pools, `std::mutex/shared_mutex/condition_variable`, TCP/IP Socket Programming, Real-Time Transaction Processing

Core Concepts: Object-Oriented Design, Operating Systems, DBMS, System Design, SDLC

Backend & APIs: Node.js, Express.js, REST APIs, JWT, Swagger/OpenAPI

Databases: PostgreSQL, SQLite, MongoDB, MySQL, Redis

Tools & Platforms: Linux/Unix, Git, GitHub Actions (CI/CD), Docker, CMake, AWS (EC2)

Testing & Performance: GoogleTest, Jest, k6 Load Testing, Latency Benchmarking (p50/p95/p99), Debugging

CERTIFICATIONS

Oracle Cloud Infrastructure 2025 Certified DevOps Professional | Oracle

Sep 2025

Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate | Oracle

Aug 2025

ACHIEVEMENTS

- Achieved a **LeetCode Contest Rating of 1694** and solved **800+ algorithmic problems** across arrays, graphs, dynamic programming, trees, and advanced data structures.
- Earned **5-Star ratings** in both C++ and SQL on HackerRank.